

Swarmbly AI — Documentation Index · Índice de documentación

One page, everything that exists, what each thing is for, and in which order to read it.

Una sola página: todo lo que existe, para qué sirve cada cosa y en qué orden leerlo.

¿Se puede citar? · May I quote it?

Antes de citar cualquier cifra, mirá [STATUS.md](#). Varios documentos de este repositorio fueron retirados o superados por una medición posterior, y siguen aquí a propósito: borrarlos dejaría el registro más limpio y menos honesto. [STATUS.md](#) dice, sin abrir nada, cuál de dos cifras es la que se sostiene. Se genera desde los documentos, así que no puede desfasarse de ellos.

Before quoting any figure, see [STATUS.md](#). Several documents here were withdrawn or superseded by a later measurement, and they stay on purpose: deleting them would leave a tidier record and a less honest one. [STATUS.md](#) says, without opening anything, which of two figures stands. It is generated from the documents, so it cannot drift from them.

Un enlace roto en este índice ya no es normal: `tests/test_docs_boundary.py` falla si un archivo rastreado cita un documento que no está en el árbol. *A broken link in this index is no longer normal: `tests/test_docs_boundary.py` fails when a tracked file cites a document that is not in the tree.*

Versions covered / Versiones cubiertas: **whitepaper v2.0** (v1.4 kept, superseded) · protocol specification **v0.2 revision 2** · reference implementation **V0** · new reference implementation **swarmbly_ref** (Ollama bench) · validation harness **swarmbly_validation**.

Start here · Empieza aquí

Read this if...	Go to
You have two minutes and want the argument	ONEPAGER_EN.md · ONEPAGER_ES.md
You have five minutes and want to know what Swarmbly is	WHITEPAPER_V2_EN.md Section 1 (or the repository root README.md)
You want the full argument for the design	WHITEPAPER_V2_EN.md (v1.4 kept as WHITEPAPER_EN.md , superseded)
You are going to implement a node or a client	SPEC_EN.md
You want to see the numbers	RESULTS_2026-09-25_refbench_EN.md for the current campaign, STATUS.md for what may be quoted, and <code>python3 swarmbly_validation/run_all.py --all</code> to recompute them
You want to know how a figure was reached, or why one was withdrawn	the research record below
You want the reasoning, the alternatives, and the open questions	WHITEPAPER_V2_EN.md Sections 5-6, 11 and 16 — where the design is argued against itself

Read this if...	Go to
You want to know where an idea came from	REFERENCES.md
You want to contribute	../CONTRIBUTING.md
You want to know who controls this	../GOVERNANCE.md

The documentation set

Whitepaper — the argument

WHITEPAPER_V2_EN.md · WHITEPAPER_V2_ES.md — v2.0, current

The case for Swarmbly, with the theory of fragmentation the earlier version lacked: the semantic unit and the fragment, the derived context budget, the five falsifiable models M1–M5 **measured against 473 real runs** (§15.9), the validation strategy (VALIDATION_STRATEGY_V2) and the fragmentation fundamentals extension (WHITEPAPER_EXT). This is the one to cite. Read it first. PDFs alongside: WHITEPAPER_V2_EN.pdf · WHITEPAPER_V2_ES.pdf. Deposited on Zenodo as [10.5281/zenodo.23031305](https://zenodo.org/record/23031305).

WHITEPAPER_EN.md · WHITEPAPER_ES.md — v1.4, superseded

Kept as the published prior-art record (Zenodo, 14 August 2026, [10.5281/zenodo.21957088](https://zenodo.org/record/21957088)). Its measurements remain valid with the corrections documented in v2.

*El planteamiento de Swarmbly, con la teoría de la fragmentación que a la versión anterior le faltaba: la unidad semántica y el fragmento, el presupuesto de contexto derivado, los cinco modelos falsables M1–M5 **medidos contra 473 corridas reales** (§15.9), la estrategia de validación (VALIDATION_STRATEGY_V2) y la extensión de fundamentos (WHITEPAPER_EXT). Éste es el que hay que citar. Se conserva v1.4 como registro de prior art publicado (Zenodo, 14 de agosto de 2026), con sus mediciones válidas bajo las correcciones documentadas en v2.*

Protocol specification — the contract

SPEC_EN.md · SPEC_ES.md — v0.2, revision 2

Normative. What an implementation must do to interoperate: message schemas, task record format, the decomposition and assembly contracts, overlap declaration, consensus rules, node obligations, error and timeout behaviour, and versioning. Written so an independent implementation can be built from it without reading the reference code — and if it cannot, that is a bug in the specification, not in the reader.

RFC 2119 keywords (MUST, SHOULD, MAY) are used and mean what they say. Changes go through the SWIP process (../CONTRIBUTING.md Section 6).

Normativa. Lo que una implementación debe hacer para interoperar: esquemas de mensajes, formato de registro de tarea, contratos de descomposición y ensamblado, declaración de solapamiento, reglas de consenso, obligaciones de los nodos, comportamiento ante errores y timeouts, y versionado. Escrita para que se pueda construir una implementación independiente sin leer el código de referencia; si no se puede, es un fallo de la especificación, no de quien la lee.

References — the bibliography

REFERENCES.md (also distributed as Swarmbly_AI_References.pdf)

The authoritative bibliography: distributed and collaborative inference, volunteer and peer-to-peer computing, task decomposition and LLM orchestration, sequence assembly and genomics, consensus in untrusted systems, and open-source governance. Every claim about prior work in the whitepaper and the master documents traces back to an entry here.

This is also the file CITATION.cff defers to for full bibliographic detail: where that file leaves a reference incomplete, the complete entry lives here.

La bibliografía de referencia: inferencia distribuida y colaborativa, computación voluntaria y P2P, descomposición de tareas y orquestación de LLM, ensamblado de secuencias y genómica, consenso en sistemas no confiables y gobernanza de software libre. Toda afirmación sobre trabajo previo en el whitepaper y en los documentos maestros se remite a una entrada de aquí.

Research record — how each figure was reached

Every measurement the whitepaper rests on has its own document, and so does every one that was withdrawn. They are kept as written; the header of each states whether it is current, superseded or withdrawn, and STATUS.md collects those states on one page. Pairs are EN · ES.

Cada medición en la que descansa el whitepaper tiene su propio documento, y también cada una que se retiró. Se conservan tal como se escribieron; el encabezado de cada uno dice si está vigente, superado o retirado, y STATUS.md reúne esos estados en una página.

Kind	Documents
Current campaign	RESULTS_2026-09-25_refbench_EN.md · _ES — 473 runs, the verdicts on M1-M5 and the L curve · VALIDATION_STRATEGY_V2_EN.md · _ES · WHITEPAPER_EXT_EN.md · _ES
Pre-registrations	PREREGISTRATION_composition.md · _ES · PREREGISTRATION_term_once.md · _ES · PREREGISTRATION_comp_final_once_EN.md · PREREGISTRATION_comp_final_once.md · PREREGISTRATION_feasibility_EN.md · PREREGISTRATION_feasibility.md · PREREGISTRATION_L_curve_EN.md · PREREGISTRATION_L_curve.md · PREREGISTRATION_2026-09-25_interaction_EN.md · _ES
Results, September 2026	RESULTS_2026-09-05_three_tiers.md · _ES · RESULTS_2026-09-05_oracle_and_k3.md · _ES · RESULTS_2026-09-05_oracle_v2.md · _ES · RESULTS_2026-09-05_term_once_EN.md · RESULTS_2026-09-05_term_once.md · RESULTS_2026-09-05_final_dedup_EN.md · RESULTS_2026-09-05_final_dedup.md · RESULTS_2026-09-22_lcurve_dev_EN.md · RESULTS_2026-09-22_lcurve_dev.md (a method defect; the L curve itself is measured in whitepaper v2, §15.4) · RESULTS_2026-09-25_interaction_EN.md · _ES · RESULTS_COMPOSITION_FINAL.md · _ES (superseded)
Results, August 2026	RESULTS_V0_V3C.md · _ES · RESULTS_V3C_FF_COMPOSITION.md · _ES · RESULTS_V3C_GT_CORRECTED.md · _ES · RESULTS_V4.md · _ES · RESULTS_V5.md · _ES · RESULTS_V6.md · _ES · RESULTS_TABLES_DEV.md · _ES · RESULTS_TABLES_DEV2.md · _ES · RESULTS_TABLES_FINAL.md · _ES (superseded) · RESULTS_TABLES_FINAL_CORRECTED.md · _ES
Reviews, incidents and decisions	REVISION_2026-08-12_EN.md · REVISION_2026-08-12.md · REVISION_2026-09-05_que_hemos_medido_EN.md · REVISION_2026-09-05_que_hemos_medido.md · STATE_2026-09-04.md · _ES · INCIDENT_2026-09-04_chain_misplan.md · _ES · FINDING_2026-09-04_segmenter_splits_question_from_data.md · _ES · POSTMORTEM_2026-09-05_four_tiers.md · _ES · ADR-001_instrument_boundary.md · _ES
Operations	RUNBOOK.md · RUNBOOK_ES.md — how to run each tier, and how to rehearse it first

Related documents outside docs/

These live in the repository root, and are listed here so this page is a complete index.

File	What it is
../README.md	Project overview and quick start
../LICENSE	The full GNU AGPL v3.0 text — verbatim, 661 lines
../NOTICE	Copyright, license summary, and the AGPL Section 13 network-use note
../CONTRIBUTING.md	EN/ES. DCO sign-off, why not a CLA, code style, the SWIP process, the measurement rule
../GOVERNANCE.md	EN/ES. Current status, the Verein → Stiftung path, what the Foundation will and will not do, and what would count as capture
../TRADEMARK.md	Trademark policy. Marks are not registered; statement of intent
../CITATION.cff	Machine-readable citation metadata (CFF 1.2.0)
../.zenodo.json	Zenodo deposition metadata
RESULTS_V0_V3C.md	First measurements against real models: the corrected coherence-tax curve, the go/no-go verdict, and the V3c agreement result

Conventions

- **Bilingual documents keep EN and ES in sync.** If you change one and cannot do the other, say so in the PR so it is tracked rather than silently drifting (../CONTRIBUTING.md Section 4).
 - **Domain vocabulary is fixed and is used precisely.** read, contig, overlap, scaffold, consensus, coverage carry their shotgun-assembly meanings throughout. Do not introduce synonyms for concepts that already have names.
 - **PDFs are generated, not authored.** The Markdown is the source of truth. A .pdf next to a .md is a build artifact; edit the .md.
 - **Version numbers are stated in each document's header** and must agree with ../CITATION.cff and ../.zenodo.json. Divergence across those three records weakens the prior-art chain: the dated record is the set of Zenodo DOIs and the v1 tag, and a title that drifts between them is a title a reader cannot follow.
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Public-facing material / Material divulgativo

- ONEPAGER_EN.md · ONEPAGER_ES.md — the two-page argument for the project, written for a general and a prospective-supporter audience: the asymmetry, why model-splitting fails, the reframing, what the first measurements showed, and what they did not show. It states the negative result rather than omitting it — a summary that hides its first failure has not earned its first success. Companion PDFs alongside.
- DIVULGACION_ES.md · DIVULGACION_EN.md — plain-language explainer of the project for a general audience. No jargon, no licensing or funding content. Companion PDFs alongside.

Note on cross-references. The v1.4 whitepaper and the specification write section references out as "Section 5.4.1" / "sección 5.4.1". Whitepaper v2 and the results documents use the section sign, "\$5.4.1".